

Sentences, Grammar & the Pipeline

Post-Assessment 2 · 12 questions · 40 points

Name: _____

Date: _____

Score: / 40

This is a **post-assessment**. It shows what you learned in this unit. Try every question, and check your answers if you have time.

Work on your own, without a computer. You have about **45 minutes**. When a question asks what a program prints, write exactly what Python would print, one line of output per answer line. If the code would crash, write the name of the error (for example, `TypeError`). Show your thinking; partial credit is given.

In every spaCy question, assume this code has already run: `import spacy` and `nlp = spacy.load("en_core_web_sm")`

1. Sentence splitter

3 points

```
text = "Ms. Park met Dr. Cruz. They waved!"
doc = nlp(text)
for sent in doc.sents:
    print(sent.text)
```

(a) Write what the loop prints.

(2 pts)

(b) How many pieces would `text.split(".")` make?

(1 pt)

2. Spans

3 points

```
doc = nlp("The old wizard opened the heavy door.")
span = doc[1:3]
```

Expression	Result
<code>span.text</code>	
<code>span.start</code>	
<code>span.end</code>	

3. Keyword Detective

4 points

Fill in the four blanks so the program prints the 5 most common keywords (as lowercase lemmas), keeping only alphabetic tokens that are not stop words.

```
from collections import Counter
doc = nlp(chapter_text)
lemmas = [t._____.lower() for t in doc if t._____ and not t.is_stop]
tally = _____(lemmas)
print(tally._____(5))
```

4. Parts of speech

4 points

spaCy reads “A tiny bird sang loudly in the old tree.” Write the `pos_` label for each word. Choose from: **NOUN**, **VERB**, **ADJ**, **ADV**.

Word	tiny	bird	sang	loudly
pos_				

5. Decoding labels

2 points

(a) What kind of word does `spacy.explain("VBD")` describe?

(1 pt)

(b) Which attribute gives simple labels like `NOUN: pos_` or `tag_`?

(1 pt)

6. Verb finder

3 points

```
doc = nlp("The kids laughed, jumped, and played in the warm rain.")
verbs = [t.text for t in doc if t.pos_ == "VERB"]
print(verbs)
```

(a) What does it print?

(2 pts)

(b) What would you change so it collects the **nouns** instead?

(1 pt)

7. Tricky sentences

2 points

spaCy tagged “I saw her duck.” like this:

SPACY OUTPUT

I PRON saw VERB her PRON duck NOUN

(a) Describe a meaning of the sentence in which “duck” should be a **VERB**. (1 pt)

(b) Why can't the model always pick the right meaning? (1 pt)

8. Keeping the spaces

4 points

```
doc = nlp("Wow, it works!")
```

(a) What does `print([t.text_with_ws for t in doc])` show? (2 pts)

(b) What does `print(" ".join([t.text for t in doc]))` show? (1 pt)

(c) Write an expression that rebuilds the original text exactly from the tokens. (1 pt)

9. Who did what?

4 points

spaCy parses “My brother kicked the ball.” Fill in the four blanks. Word bank for `dep_`: **nsubj**, **dobj**, **ROOT**, **poss**.

Token	dep_	head . text
My	poss	
brother		kicked
kicked		kicked
ball		kicked

10. Sentence Skeletons

2 points

(a) Write the import line you need before you can call `displacy.render(...)`. (1 pt)

(b) In “My brother kicked the ball.”, list the `children` of the token “kicked”. (1 pt)

11. The assembly line

4 points

OUTPUT OF `PRINT(NLP.PIPE_NAMES)`

```
['tok2vec', 'tagger', 'parser', 'attribute_ruler', 'lemmatizer', 'ner']
```

(a) Which component adds lemmas (`lemma_`)? (1 pt)

(b) Which component predicts the detailed tags (`tag_`)? (1 pt)

(c) Why turn off components you don't need? (1 pt)

(d) Write a line that loads `"en_core_web_sm"` with the `parser` component turned off. (1 pt)

12. Most common adjectives

5 points

Write a function `top_adjectives(texts, n)` that takes a **list** of texts, processes them with `nlp.pipe`, and **returns** the `n` most common **ADJ lemmas** in lowercase (use `Counter`).

EXAMPLE: `TOP_ADJECTIVES(["A BIG DOG MET A SMALL CAT.", "THE BIG HOUSE HAD A RED DOOR.", "BIG DREAMS!"], 1)` RETURNS

```
[('big', 3)]
```

END OF ASSESSMENT